



NASA's DART Proves It's Serious About Planetary Defense

Deflecting asteroids, defending Earth!

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The asteroid hurtled along its path, shedding orange embers as it made its way to its target. The darkness it had emerged from created a silent veil that absorbed all light until all that existed was the giant rock and anything that lay in its path. It moved with purpose, gaining speed until it finally sunk against the cold, metallic spacecraft hailing from a little planet called Earth.

No, this isn't an excerpt from a Sci-Fi novel - it's what happened on September 26, 2022, when NASA's spacecraft crashed into Dimorphos, an asteroid that was a part of a double-asteroid system. The

collision was carried out as a part of NASA's Double Asteroid Redirection Test (DART), an effort to determine whether large asteroids moving toward Earth could be redirected. You know, to save humanity.

WHY DID WE FLING A SPACECRAFT AT A RANDOM ASTEROID?

The idea of defending the Earth from life-threatening asteroids had been doing the rounds at NASA and the European Space Agency for a while. The two organisations joined forces with other international entities in 2015 to form the Asteroid Impact and Detection Assessment (AIDA). The primary

function of AIDA is to help create defense protocols for Earth in the event that a large asteroid threatens its being. AIDA efforts comprise DART (from NASA), LICIACube (from Agenzia Spaziale Italiana) and HERA (from ESA).

As the name indicates, DART's primary function is to test whether crashing a spacecraft into an asteroid is a feasible solution for veering it off course. It is the first ever test of its kind, and has been created to demonstrate NASA's kinetic impactor technology (which relies on the notion that kinetic impact can be used to veer an asteroid off its course in space).

Scientists chose Dimorphos, a small asteroid orbiting a larger one known as Didymos. This asteroid is estimated to have a mass of 5 billion kilograms, and will never intersect Earth's orbit, which means it was never a threat to begin with. Additionally, its impact with the 560kg DART spacecraft will also

